Financial Calculators: Test Cases

How to use these

Today we're going to be Quality Assurance Testers (QA Testers) and test our program before putting it in front of a customer. Type the numbers from each row into your calculator. If your answer matches the Expected column, your calculator works.

A penny off is okay. Java can't store most decimal numbers perfectly, so your answer might be off by a cent or two. That's normal. Use double for your numbers, not float. If you're off by whole dollars, something else is wrong, so check your formula.

Calculator 1: Mortgage

Principal	Interest rate	Years	Expected monthly payment	Expected total interest
$53,000	7.625%	15	$495.09	$36,115.99
$200,000	6.5%	30	$1,264.14	$255,088.98
$150,000	3%	15	$1,035.87	$36,457.04
$25,000	5%	5	$471.78	$3,306.85

Calculator 2: CD future value

Deposit	Interest rate	Years	Expected future value	Expected interest earned
$1,000	1.75%	5	$1,091.44	$91.44
$10,000	5%	1	$10,512.67	$512.67
$2,500	2.25%	3	$2,674.57	$174.57
$5,000	4%	10	$7,458.96	$2,458.96

Note on case 1. The workbook lists $1,092.62 for this example, but its own daily-compounding formula gives $1,091.44. If you get $1,091.44, your code is correct.

Tip: 1.75% needs to become 0.0175 in your code. Divide the rate by 100.

Calculator 3: Annuity present value

Monthly payout	Interest rate	Years	Expected present value
$3,000	2.5%	20	$566,141.46
$1,000	5%	10	$94,281.35
$500	6%	5	$25,862.78